MIL-R-43650

20 June 1969

(see 6.3)

MILITARY SPECIFICATION

RATION SUPPLEMENT, BEVERAGE PACK

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers components and packaging requirements for beverage supplements (beverage base powder, instant tea, lemon flavored tea, and refined sugar), for issue to Armed Forces personnel in conjunction with operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

JJJ-S-791 - Sugar, Refined and Brown; Beet or Cane.
PPP-B-26 - Bag, Plastic, Polyethylene.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.

MILITARY

MIL-T-3527 - Tea, Instant.

STANDARDS

FEDERAL

FED-STD-101 - Preservation, Packaging and Packing Materials: Test Procedures

FSC 8970

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Plants.

(Copies of specifications and standards, required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

U.S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

The United States Pharmacopoeial Convention , Inc.

Pharmacopoeia of the United States.

(Application for copies should be addressed to the United States Pharmacopoeial Convention, Inc., 46 Park Avenue, New York, NY 10016.)

National Research Council

Food Chemicals Codex.

(Application for copies should be addressed to the Office of Director, National Academy of Sciences, National Research Council, Washington, DC 20418.)

U. S. Army Natick Laboratories

Food Technology - Hedonic Scale Method of Measuring Food Preferences.

(Application for reprints should be addressed to the U.S. Army Natick Laboratories, ATTN: AMXRE-FS, Natick, MA 01760.)

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:56Z
Check the source to verify that this is the current version before use.

Association of Vitamin Chemists

Methods of Vitamin Assay, 3rd Edition.

(Application for copies should be addressed to the Interscience Publishers, a Division of John Wiley and Sons, Inc., 605 Third Avenue, New York, NY 10016.)

Association of Official Agricultural Chemists

Official Methods of Analysis.

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, DC 20044.)

3. REQUIREMENTS

3.1 Bid sample approval (beverage base powder and lemon flavored instant tea with sugar added). Twelve (unless otherwise specified, see 6.1), individually packaged samples, each containing a sufficient quantity of beverage base powder, with full complement of sugar, to make 12 ounces of beverage of each flavor, of equal and uniform quality, representative of the product which the bidder proposes to furnish under the contract, shall be submitted to the contracting officer prior to the bid opening date. Four samples of each flavor will be used by the contracting officer to determine bid sample approval by panel testing (see 4.6). The remaining 8 samples of each flavor submitted by the bidder shall be used as standards for determining compliance with the palatability requirements. The approval of any bid sample for palatability will not constitute approval of the sample as meeting the other requirements of this specification.

3.2 Ingredients of beverage base powder.

3.2.1 Flavors.- Flavors shall be natural or imitation and shall be entrapped or encapsulated by incorporating them in vegetable gum, sorbitol, dextrose, or other edible material, in such a manner that the flavors are protected from oxidation, rancidity, evaporation, and other types of deterioration.

3.2.2 Oil of orange.- Oil of orange shall be United States Pharmacopoeia cold-pressed orange oil of single strength or partially deterpenized oil prepared therefrom, or combinations of these.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

3.2.3 Oil of lemon.- Oil of lemon shall be USP cold-pressed lemon oil of single strength or partially deterpenized oil prepared therefrom, or combinations of these.

3.2.4 Oil of lime.- Oil of lime shall be high-quality distilled, or cold-pressed oil of lime of single strength or partially deterpenized oil prepared therefrom, or combinations of these.

3.2.5 Acids.- Food acidulants shall conform to the requirements specified in Food Chemical Codex. Fumaric acid, when used, shall be treated to insure rapid solubility in cold water.

3.2.6 Sugar.- Sugar shall be white, refined, granulated cane or beet sugar or a combination thereof.

3.2.7 Dextrose.- Dextrose shall be dextrose hydrate.

3.2.8 Coloring materials.- Coloring materials shall be vegetable colors or approved U.S. certified food colors. They shall be acid resistant.

3.2.9 Ascorbic acid.- Ascorbic acid shall conform to the requirements of Food Chemical Codex.

3.2.10 Emulsifying, coating, clouding and entrapping or encapsulating agents.- Emulsifying, coating, clouding and entrapping or encapsulating agents shall be non-deleterious and of the types commonly used in edible products.

3.3 Beverage base powder (finished powder) and prepared beverage.-

3.3.1 Beverage base powder.- Beverage base powder shall be a mixture of acids, flavors, ascorbic acid, and coloring materials, with a full complement of sugar, and shall comply with 3.3.1.1 through 3.3.1.3. Optional ingredients, such as antioxidants or emulsifying, coating, anti-caking and entrapping agents, and carriers or fillers, such as dextrose or lactose, may be included in the mixture. The beverage base powder shall be of the following flavors: Orange, lemon, lime, grape and cherry.

3.3.1.1 Moisture content.- Moisture content of beverage base powder shall be not more than 7.0 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

3.3.1.2 Ascorbic acid.- The beverage base powder shall contain an added amount of ascorbic acid of not less than 20 milligrams (mg.) nor more than 30 mg. per quantity of powder required for 12 fluid ounces of finished beverage.

3.3.1.3 Physical condition.- The beverage base powder shall be free flowing and readily soluble (or dispersible) in cold water.

3.3.2 Prepared beverage.- Sufficient quantity of beverage base powder shall be used to make 12 fluid ounces of beverage (see 5.1.1). The resulting beverages shall possess characteristic, bright colors, typical of the respective freshly-extracted fruit juices of the specified flavors. The beverages shall possess palatability (distinct flavor, aroma, mouth-feel, etc.) reasonably characteristic of the freshly-extracted fruit juice of the flavor specified, and shall be free from oxidized, terebinthic, and other undesirable flavors. The finished beverages shall show no coagulation or gelation, and shall be free from foreign materials.

3.3.2.1 Palatability.- The palatability of the beverage prepared from delivery samples shall be equal to or better than the beverage prepared from the bid sample submitted by the successful bidder.

3.4 Tea, instant (for tea envelope).- Instant tea shall conform to the requirements for type II of MIL-T-3527.

3.5 Sugar, refined (for sugar envelope).- Sugar, refined, shall conform to Type I (a) of JJJ-S-791.

3.6 Tea, instant (lemon flavored with sugar).- Tea instant (lemon flavored with sugar) shall be a mixture of instant tea, acid, lemon flavor with a full complement of sugar and shall comply with applicable ingredients specified in 3.2. Optional ingredients such as, spray dried lemon peel oil, malto-dextrin, antioxidants, BHA (used as preservative) may be used. In addition, it shall comply with requirements in 3.3.1.3, 3.3.2.1, and 3.6.1.

3.6.1 Moisture content.- Moisture content of tea, instant (lemon flavored with sugar) shall be not more than 1.0 percent.

3.7 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated thereunder.

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:56Z
Check the source to verify that this is the current version before use.

3.8 Workmanship.- The product shall be prepared, processed, and packaged under modern sanitary conditions and in accordance with good commercial practice. The product shall be processed in establishments meeting the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preacceptance inspection.- Product shall be rejected if produced in plants not meeting the requirements of MIL-STD-668.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specification, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Examination of components.- Examination of components and ingredients described in this specification shall be in accordance with 4.3.1.1.1 and tables II through VI. Sampling plans and acceptance criteria for the tables of examination shall be in accordance with table I. The acceptable quality levels (AQLs) shall be expressed as percent defective for table IV, and defects per hundred units for tables II, III, V and VI.

4.3.1.1.1 Ingredients of beverage base powder and lemon flavored instant tea (with added sugar).- Conformance of ingredients to identity requirements (kinds, strength, USP, Food Chemical Codex, etc., as applicable), shall be ascertained by examination of invoices, labels, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to 3.2. The sample unit shall be approximately 4 ounces of each ingredient extracted from a container. The sample size shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

be 5 sample units. In the event a lot of ingredients consists of less than 5 containers, a sample unit shall be extracted and examined from each container in the lot. Nonconformance to identity requirements, or one or more sample units not in conformance to condition requirements, shall indicate an unacceptable component and use of such shall be cause for rejection of the involved quantity of beverage base.

TABLE I.- Sampling plans and AQLs.

Table	Inspection Level	Sample Unit	Lot Size expressed in	AQLs Major	AQLs Minor	AQLs Total
II 1/	S-4	1 filled case of beverage base or lemon flavored tea envelopes	Cases	1.5	---	4.0
III	I	1 filled and sealed beverage base or lemon flavored tea envelope	Envelopes	.65	---	2.5
IV	S-3	1 filled and sealed beverage base or lemon flavored tea envelope	Envelopes	---	1.5	---
V	S-3	1 filled and sealed beverage base or lemon flavored tea envelope	Envelopes	1.0	---	---
VI 2/	S-2	1 filled and sealed beverage base or lemon flavored tea envelope	Envelopes	1.5	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

1/ Applicable only to beverage base and lemon flavored tea which are Government furnished property.

2/ Each flavor shall be examined as a separate lot.

TABLE II.- Examination of cases containing beverage base and lemon flavored tea

Category		Defect
Major	Minor	
101		Markings missing, incorrect, or illegible.
	201	Marking for net weight indicates a net weight greater than maximum specified.
	202	More than 2 percent less envelopes than marked number. 1/

1/ Lot shall be rejected if sample data indicate lot average quantity of envelopes per case less than marked number.

TABLE III.- Examination of beverage base and lemon flavored tea envelopes

Category		Defect
Major	Minor	
101		Tear, hole, sifter, or open seal.
102		Not type or size specified. 1/
103		Delamination.
104		Not clean.
	201	Objectionable odor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

TABLE III.- Examination of beverage base and lemon flavored tea envelopes (Contd)

Category		Defect
Major	Minor	
	202	Color not as specified
105		Nomenclature or directions for use missing, incorrect, or illegible.
	203	Other required label information missing, incorrect, or illegible.
	204	Seal width less than 1/4 inch at any point. 1/

1/ Measurements shall be made with a device calibrated in 1/16 inch increments.

TABLE IV.- Examination for net weight of beverage base and lemon flavored tea 1/

Category	Defect
Minor	
201	Less than 32.3 grams (for beverage base envelopes)
202	Less than 34.2 grams (for lemon flavored tea)

1/ Report results to nearest 0.1 gram. Lot shall be rejected if sample data indicate lot average net weight of less than specified net weight.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

TABLE V.- Examination for leakage 1/ and delamination 2/ of beverage base and lemon flavored tea envelopes

Category	Defect
Major	
101	Leakage.
102	Delamination.

1/ See 4.5.1.1 for procedure.

2/ Examine for delamination after removing envelopes from water.

TABLE VI.- Examination of beverage base and lemon flavored tea (powdered and prepared 1/

Category	Defect
Major	
	Powdered
101	Not free flowing.
	Prepared 2/
102	Produced not readily soluble or particles do not disperse readily.
103	Color of the beverage not a facsimile of the fruit represented (as applicable).
104	Coagulation or gelation (as applicable).
105	Palatability not equal to or better than bid sample

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

1/ Presence of foreign material (e.g., glass, wood, paint, metal, filth, insects, etc.) shall be cause for rejection of the lot.

2/ See 5.4.1 and 5.4.1.1 for preparation. Utilize contents of filled envelope of bid sample to prepare standard beverage or standard lemon flavored tea.

4.3.1.2 Testing of beverage base, lemon flavored tea and related envelope material.- Sampling plans for testing shall be in accordance with table VII. Test requirements shall be average requirements. A test failure shall indicate an unacceptable component and use of such shall be cause for rejection of the involved quantity of finished beverage base or lemon flavored tea envelopes.

TABLE VII.- Testing of beverage base, lemon flavored tea and related envelope material

Component	Sample Unit	Lot size Expressed in	Inspection level	Characteristics	Results Reported to nearest	Test ref & reqmt par
Envelopes	1 yd. of material or equivalent in envelopes	Envelopes or rolls	S-2 S-4	Thickness of 1/		
				Polyethylene	0.0001 inch	5.1.1
				Aluminum	0.00001 inch	5.2.2
				Basis wt. of paper	lb/ream	4.5.2.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

TABLE VII.- Testing of beverage base, lemon flavored tea and related envelope material (Con'd)

Component	Sample Unit	Lot size Expressed in	Inspection level	Characteristics	Results Reported to nearest	Test ref & reqmt par
Beverage base and lemon flavored tea.	2/	Envelopes	S-2	Moisture	0.1 percent	3.3.1.1 3.6.1 4.5.2.1.1
				Ascorbic acid (beverage base powder only)	Mg.	3.3.1.2 4.5.2.1.2

1/ In lieu of performing thickness tests on laminae, compliance with thickness requirements may be ascertained by examination of pertinent labels, invoices or other valid documents.

2/ The sample shall be a composite formed from the number of envelopes indicated by the inspection level. The composite shall be formed at the laboratory just prior to testing. Each flavor shall be considered a separate lot. A 4 ounce sample of unfortified product representing the lot prior to fortification shall accompany the sample of finished beverage powder.

4.3.2 Examination of assembly.- Examination of assembly shall be in accordance with tables IX through XI and 4.3.2.1 and 4.3.2.2. Sampling plans and acceptance criteria shall be in accordance with table VIII. the AQLs shall be expressed as defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 202640-21T21:56Z
Check the source to verify that this is the currentversion before use.

TABLE VIII.- Sampling plans and AQLs for assembly

Table	Inspection level	Sample Unit	Lot size express in	AQLs Major	AQLs Total
IX	S-3	1 filled bag contents (lemon flavored tea and tea and sugar envelopes)	Bags	6.5	10.0
X	S-3	1 filled carton of beverage base envelopes	Cartons	6.5	10.0
XI	S-3	1 filled intermediate container	Intermediate containers	1.0	4.0

TABLE IX.- Examination of contents of bags containing lemon flavored tea and tea and sugar

Category		Defect
Major	Minor	
101		Less than 15 tea envelopes.
102		Less than 15 sugar envelopes.
103		Less than 5 lemon flavored tea envelopes.
	201	More than 15 tea envelopes. 1/
	202	More than 15 sugar envelopes. 1/
	203	More than 5 lemon flavored tea envelopes. 1/

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

TABLE IX.- Examination of contents of bags containing lemon flavored tea and tea and sugar (Cont'd)

Category		Defect
Major	Minor	
104		Tear, hole or open seal in an envelope.
105		Not clean.

1/ Defect applicable only if the components are Government furnished.

TABLE X.- Examination of beverage base carton contents

Category		Defect
Major	Minor	
101		Less than 45 envelopes.
	201	More than 45 envelopes. 1/
	202	Less than 9 envelopes of each flavor.
102		Tear, hole, or open seal in an envelope.
103		Not clean.

1/ Defect applicable only if beverage base envelopes are Government furnished.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

TABLE XI.- Examination of intermediate pack contents

Category		Defect
Major	Minor	
101		Carton missing.
102		Polyethylene bag missing.
	201	Placement of polyethylene bag not as specified.
	202	Polyethylene bag not closed and secured as specified, or has tear, hole, or open seam greater than 1 inch in any dimension. (If heat sealed, excess air in bag resulting in tight fit for available space).
	203	Carton not securely closed or has tear or hole greater than 2 inches in any dimension.
	204	Label information on carton missing, incorrect, or illegible.

4.3.2.1 Examination of intermediate containers for closure and marking.- Examination shall be in accordance with the criteria in the appendix of PPP-B-636. In addition, the following defect shall be included in the examination: Major - Markings missing, incorrect, or illegible.

4.3.2.2 Examination of shipping container for closure, reinforcement and marking.- Examination shall be in accordance with the criteria in the appendix of PPP-B-636. In addition, the following defect shall be included in the examination: Major - (1) Markings missing, incorrect, or illegible; (2) Intermediate container missing.

4.4 Standby test samples.- The Government reserves the right to withdraw and hold standby test samples of components or finished product or both (quantity of which shall be not more than twice that required by this specification) for inspection purposes. Samples not used will be returned to the supplier, upon request.

Source: https://assist.dla.mil -- Downloaded: 20260-21T21:56Z .
Check the source to verify that this is the current version before use.

4.5 Test methods.- Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

4.5.1 Examination procedures.

4.5.1.1 Leakage.- The filled and sealed envelope shall be examined for leakage by submerging in water contained in a desiccator or other suitable container, and maintaining a vacuum of 10 inches of mercury (atmospheric pressure 29.9 inches) for at least 30 seconds. A leak is indicated by a steady progression of bubbles. Isolated bubbles caused by entrapped air are not considered as signs of leakage.

4.5.2 Test procedures.

4.5.2.1 Chemical analyses (beverage base powder and lemon flavored instant tea with sugar added.-

4.5.2.1.1 Moisture. Chemical analysis for moisture shall be made in accordance with the method described in Official Methods of Analysis of the Association of Official Agricultural Chemists; chapter: Sugar and Sugar Products; section: Sugars, Sirups and Molasses; method: Moisture (Direct Drying).

4.5.2.1.2 Vitamin determination.- Ascorbic acid content shall be determined by the method described in Methods of Vitamin Assay, 3rd edition, chapter: Ascorbic acid, pages 304(3) to 306. The amount of ascorbic acid in the fortified sample shall be calculated by difference in the assay values of the unfortified and fortified samples, taken from the same lot.

4.5.2.2 Basic weight of paper test.- Basic weight of paper shall be tested in accordance with method 204 of FED-STD-101.

4.6 Palatability.- Tests for palatability, as required, shall be performed in accordance with 4.6.1 through 4.6.3, as applicable. Each flavor shall be considered a separate product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

4.6.1 Procedure.- Samples shall be prepared as described in 3.3.2. Approximately 2-ounce portions shall be served, at a temperature of 50°F. ±2°F., in coded glass containers, to a consumer-type panel. The panel members shall rate each sample on a 9-point hedonic scale*, graduated in successive degrees of like and dislike. The samples shall be presented successively with a minimum interval of 30 seconds between samples in controller orders, so that each sample is served an equal number of times in each temporal position. Each subject shall rate from 2 to 6 samples at a test session, but all subjects shall rate the same number of samples in a given session. The subjects shall test without interference from each other or from outsiders.

4.6.2 Selection of reference standards.- Samples complying with the requirements of this specification will be tested at the same session by a consumer-type panel (yielding not less than 40 ratings per sample). Tests will be conducted according to 4.6.1. Samples which are acceptable and can be used for reference standards shall be those samples that are not significantly different from the highest ranking sample at the 5-percent level. Reference samples shall be stored in closed containers and held at a temperature of approximately 40°F. The reference standards for any given test session shall be selected randomly from among the available reference standards.

* 9	8	7	6	5	4	3	2	1
Like extremely	Like very much	Like moderately	Like slightly	Neither like nor dislike	Dislike slightly	Dislike moderately	Dislike very much	Dislike extremely

NOTE: For further details, see Hedonic Scale Method of Measuring Food Preferences, reprinted from Food Technology, 1957, Vol. XI, No. 9, Supplement pages 9-14.

4.6.3 Bid sample approval.- Bid samples shall be tested along with reference standards as prescribed in 4.6.1. A minimum of 24 responses shall be obtained for each bid sample. One or more reference standards shall be served in each pre-award session. Any bid sample which has a mean rating significantly lower, at the 5-percent level, than that of the referenced sample, shall be rejected. Approved samples shall be held in closed containers at approximately 40°F. to serve as standards for testing of deliveries.

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:56Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- The components shall be packaged in accordance with 5.1.1, 5.1.2 and 5.1.3, as applicable.

5.1.1 Beverage base powder.- A net weight of 34.0 grams of the product shall be packaged in a heat-sealable envelope having inside dimension of 3 7/8 x 4 1/4 inches ± 1/4 inch in either direction. All 3 seams and the closure shall be made by heat sealing with seals having a width of not less than 1/4 inch. The envelope material shall be made by laminating or hot extruding 0.0015 inch thick polyethylene onto a 0.00035 inch thick aluminum foil, and laminating the plain side of the foil to sulphite or kraft paper. The paper shall be bonded to the aluminum foil by hot extruding 8 pounds per ream of polyethylene between the layers. The paper shall have a basis weight of not less than 25 pounds per ream (24 x 36-500) when tested as specified in 4.5.2.2, and shall be treated or processed to form a soft pliable sheet. The combined sheet shall show no evidence of delamination when made into envelopes, tested for leakage, or heat sealed. The outside of the envelopes shall be natural kraft or similar dull brown color. Envelopes shall show no leakage when tested in accordance with 4.5.1.1.

5.1.2 Instant tea.- A net weight of 0.65 gram of type II product shall be packaged in an envelope in accordance with MIL-T-3527.

5.1.2.1 Tea, instant (lemon flavored with sugar).- A net weight of 36 grams of the product shall be packaged in a heat-sealable envelope in accordance with 5.1.1.

5.1.3 Refined sugar.- A net weight of 6 grams of type I (a) product shall be packaged in an envelope in accordance with JJJ-S-791.

5.2 Assembly.- The components, packaged as specified in 5.1, shall be assembled in accordance with 5.2.1 and 5.2.2 into an intermediate container specified in 5.2.3.

5.2.1 Beverage base envelopes.- Forty-five envelopes (9 of each flavor specified in 3.3.1) shall be stacked flat or on end in a folding paperboard box complying with one of the following styles of PPP-B-566: Style V; Style VIII, type A, class b; Style X or Style XI. Boxes shall be constructed from

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

not less than 0.030 inch thick paperboardcomplying with variety 1 of PPP-B-566. The paperboard shall be natural kraft in color. The top lid of the box shall be equipped with tuck locks or pye locks which shall be in the locked position when the lid is closed. The box shall have inside dimensions of 12 by 4-3/4 by 5 inches.

5.2.2 Tea and sugar envelopes.- Fifteen tea envelopes, 5 lemon flavored tea envelopes, and 15 sugar envelopes, packaged in accordance with 5.1.2, 5.1.2.1 and 5.1.3, respectively, shall be filled into a polyethylene bag complying with type II of PPP-B-26. The bag shall measure 7 by 12 inches, and the polyethylene shall be not less than 0.002-inch thick. The envelopes shall be spread out within the bag so as to form a flat package that will fit into the available space. The polyethylene bag shall be closed by folding the top over against the body of the bag by the application of 1-inch wide pressure sensitive cellophane tape that overlaps each of the adjoining surfaces at least one inch. Alternatively the bag may be heat sealed provided sufficient air is expelled from the bag so it will fit in the available space. The polyethylene bag containing the envelopes shall be inserted on its side in the space between the ends of the paperboard boxes and the end of the intermediate container.

5.2.3 Intermediate containers.- Two paperboard boxes packaged as specified in 5.2.1 and one bag packaged as specified in 5.2.2 shall be packaged into an intermediate container. The intermediate container shall be a fiberboard box made from B-flute board constructed and closed by gluing in accordance with style RSC W5c of PPP-B-636. The intermediate container shall have inside dimensions of 13 1/4 by 9-5/8 by 5-1/8 inches.

5.3 Packing.- The completed ration shall be packed in accordance with 5.3.1. When components are shipped to the ration assembler as Government furnished property, the components shall be packed in accordance with 5.3.2.

5.3.1 Level A.- Two filled and closed intermediate containers shall be inserted into a shipping container constructed and closed in accordance with style RSC V2s of PPP-B-636. The container shall be metal strapped in accordance with PPP-B-636 except that two girthwise straps shall be applied trisecting the 13-3/4 inch dimension. The shipping container shall have inside dimensions of 13-3/4 x 5-3/4 x 20 inches.

5.3.2 Level C.- Not more than 30 pounds of the beverage base and lemon flavored tea components packaged in accordance with 5.1.1 shall be packed into fiberboard containers which are acceptable by common carriers for safe

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

transportation to the point of destination specified in the shipping instructions at the lowest transportation rate for such supplies. Tea and sugar packets shall be shipped to the place of ration assembly in accordance with MIL-T-3527 and JJJ-S-791, respectively.

5.4 Labeling and marking.-

5.4.1 Beverage base envelopes.- Envelopes shall be printed in black, bold-faced type with an indelible ink. The envelopes shall contain the following information:

Beverage Base Powder
Flavor (as applicable)
With vitamin C added
Net weight - 1.2 oz.
Name and address of manufacturer

Directions

Dissolve contents of this envelope with 12 ounces (1/2 canteen cup) of cold water.

Allow water just chemically purified to stand 30 minutes before adding beverage powder.

5.4.1.1 Tea, instant (lemon flavored with sugar).- Envelopes shall be printed in accordance with 5.4.1 with the following information:

Lemon Flavored Tea with Sugar Added
Net weight 36 Grams
Name and address of manufacturer

Directions

Dissolve contents of this envelope with 12 ounces (1/2 canteen cup) of cold water.

Allow water just chemically purified to stand 30 minutes before adding beverage powder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

5.4.2 Paperboard boxes.- The following information shall be printed or hand stamped on the boxes in black bold-faced type with an indelible ink:

Beverage Base Envelopes
5 assorted flavors - 9 each

5.4.3 Intermediate container.- Intermediate containers shall be marked in accordance with the requirements for interior packages of MIL-STD-129 and shall contain the following specific information:

Ration Supplement, Beverage Pack
Serves 100 Men
90 Beverage Powder
15 Instant Tea
5 Lemon Flavored Tea
15 Sugar

5.4.4 Shipping container.- All shipping containers shall be marked in accordance with MIL-STD-129. Containers filled with the ration supplement, beverage pack, shall also include the following information:

Ration Supplement, Beverage Pack
2 units - 100 servings each

In addition, product nomenclature shall appear on both ends and on the container.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When other than 12 samples of the beverage base powder is required (see 3.1)

6.2 It is recommended that the paperboard box be placed on end and the envelopes be assembled flat. This procedure will assist in obtaining an even distribution of powder within the envelope and will contribute to ease of assembly by reducing the space necessary to accommodate 45 envelopes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

6.3 Supersession data.- This specification includes the requirements of Interim Purchase Description IP/DES S-21-7 dated 5 June 1967.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Preparing activity:

Army - GL

Project No. 8970-0079

Review activities:

Army - MD
Navy - MC, SA
Air Force - 45
DSA - SS

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-R-43650 Ration Supplement, Beverage Pack

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:56Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01760

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-BQS
Natick, Massachusetts 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.